End-to-End Retail & Sales Inventory Optimization Project Report

1. Executive Summary

This project delivers a comprehensive, end-to-end data pipeline and analytics solution for retail and inventory optimization. Handling a large-scale dataset of **301,647 rows and 31 columns**, the project bridges raw data engineering, relational database schema design, and executive-level business intelligence visualization.

Key performance metrics achieved by the organization:

- **Total Revenue:** ~$76.78M
- **Total Unique Customers:** 86,728
- **Total Inventory Purchases:** 1.61M+ units
- **Average Order Value (AOV):** $255.16

2. Project Architecture & Tech Stack

The project follows a modern three-tier data architecture:

1. **Data Processing & Cleaning (Python):** Handling missing values, date format standardization, whitespace trimming, and duplicate checks.
2. **Database Management & Schema Design (MySQL):** DDL script creation, relational table definition, and robust data modeling.
3. **Data Visualization & Analytics (Power BI):** Interactive executive dashboard featuring DAX measures, dynamic filtering, KPI cards, trend analysis, and category breakdowns.

3. Data Engineering & Cleaning (Python Phase)

Before loading into storage layers, raw transactional logs frequently contain anomalies. Python (Pandas) was deployed to sanitize the dataset:

- Standardized date-time formats to ensure temporal accuracy across monthly and yearly trends.
- Addressed missing category fields and normalized string variables to prevent grouping fragmentation in BI tools.
- Verified row and column integrity, finalizing the clean asset (Cleaned_Retail_Sales.csv).

4. Database Schema & Architecture (MySQL Phase)

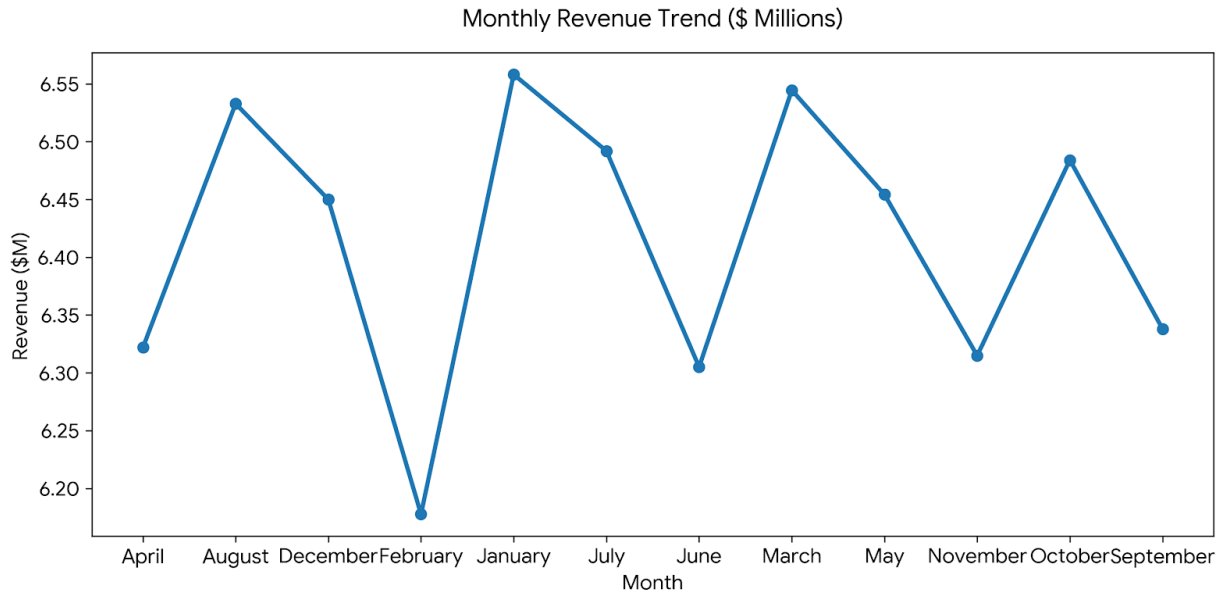
To demonstrate data warehousing capabilities, a relational schema (retail_inventory_db) was structured using SQL:

- **DDL Implementation:** Explicit column typing (INT, VARCHAR, DECIMAL) optimized for analytical query performance.
- **Integrity & Indexing:** Ensured primary keys for transactions and foreign key readiness for scalable dimensional reporting.

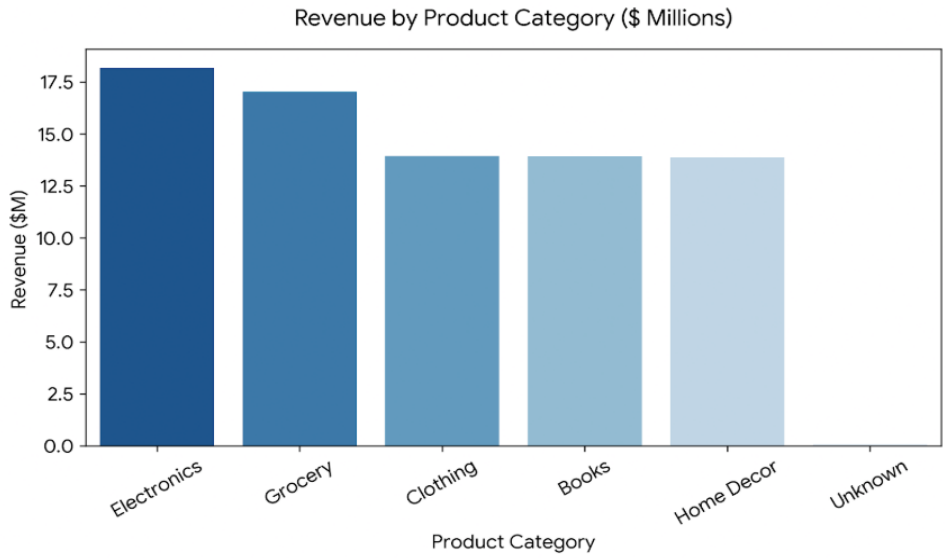
5. Business Intelligence & Dashboard (Power BI Phase)

The finalized Power BI dashboard was built to communicate actionable insights to stakeholders:

- **KPI Cards:** Displaying real-time aggregated metrics (Total Revenue, Total Customers, Total Purchases, and Avg Order Value).
- **Monthly Revenue Trend:** Line chart tracking revenue fluctuations over twelve months.



- **Revenue by Category:** Donut chart illustrating proportional revenue generation across departments (Electronics, Grocery, Clothing, Books, Home Decor).



- **Top 10 Products by Revenue:** Bar chart highlighting top-performing merchandise.
- **Interactive Filtering:** Global slicers for Country and Payment Method to drill down into specific regional and transactional behaviors.

6. Conclusion & Business Recommendations:-

- **Inventory Balancing:** High-performing categories like Electronics and Grocery drive the bulk of top-line revenue; inventory reorder points should be tightened for these categories to prevent stockouts.
- **Payment Preferences:** Digital payment channels dominate, highlighting the need for seamless checkout gateways.
- **Portfolio Impact:** This project showcases advanced proficiency in Python data pipelines, relational database design, and enterprise-grade Power BI visualization.